



MANIPAL INSTITUTE OF TECHNOLOGY
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ASSIGNMENT PROJECT REPORT

ASSIGNMENT 6

for

ECE3129: VLSI Fabrication

“Describe the Gate replacement Process”

Submitted by

NAME: Shriman Narayan Singh – 230959064 - Roll no-18

Devjeet Nanda – 230959068– Roll no-19

Arnav Sharma – 230959070 – Roll no-20

Raghunandhan PS - 230959072 - Roll no – 21

Varadharaj M Bhat – 230959074 - Roll no - 22

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OBJECTIVE:

The main objective of the Replacement Metal Gate (Gate-Last) process is to form a high-performance, thermally stable metal gate structure that overcomes the limitations of the traditional polysilicon gate while maintaining compatibility with high-k dielectrics and advanced CMOS scaling.

INTRODUCTION

The Replacement Metal Gate (RMG) or Gate-Last process is a key step in modern CMOS fabrication, introduced to overcome issues in the traditional *polysilicon gate-first* approach—mainly Fermi-level pinning, poly depletion, and gate dielectric degradation. In the gate-last scheme, the gate electrode is *not finalized* until after high-temperature steps (such as source/drain activation anneals). Instead, a dummy gate made of polysilicon is used temporarily to define the gate geometry and protect the channel during front-end processing.

Importance of the Gate-Last Approach

- **Thermal stability:** The metal gate is added *after* high-temperature steps, preventing metal diffusion or reaction with the dielectric.
- **Work function tuning:** Different metal combinations (n-type vs. p-type gates) allow threshold voltage control.
- **Scalability:** Compatible with high-k dielectrics and FinFET/advanced nodes (<22 nm)

Metal diffusion (e.g., Aluminium) into the active region can drastically degrade device performance or even cause failure by:

- Introducing trap states or leakage paths in the channel.
- Shifting the threshold voltage.
- Reducing carrier mobility.

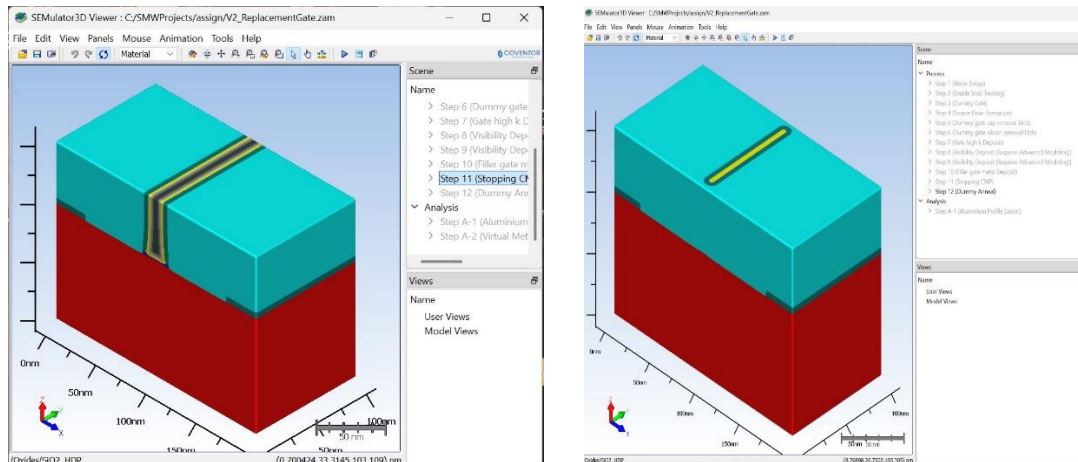
If the metal concentration in the active region exceeds 10^{12} cm^{-3} , the device is considered non-functional.

REVIEW QUESTIONS

1. Please open V2_ReplacementGate.zam. Look at the cross-section. Look at the process steps. Notice that it consists of a gate last process example as the gate metal stack is deposited AFTER the source and drain formation. Please note that the device stops working if the metal contamination exceeds 1012 cm⁻³.

- a. The polysilicon dummy gate is refractory, so does not diffuse. Al diffusion only occurs during subsequent high-temperature steps. We have added a dummy anneal at the end to represent the combined effect of all anneal steps. Plot the metal diffusion in the structures after the dummy anneal. What is the Al concentration in the active region? Use the 'Aluminum Profile Export' analysis result to compute the average concentration in the active region. Is the device working?

Answer:



Steps in the Replacement Metal Gate (Gate-Last) Process

1. **Dummy Gate Formation:**
 - Deposit and pattern a **sacrificial polysilicon gate** over a high-k gate dielectric.
 - This dummy gate defines the transistor channel length and protects the underlying dielectric during early processing.
2. **Spacer and Source/Drain Formation:**
 - Form **sidewall spacers** around the dummy gate.
 - Perform **source/drain implantation and annealing** to activate dopants and create junctions.
 - The dummy gate remains intact during these high-temperature steps to prevent metal degradation.
3. **Interlayer Dielectric (ILD) Deposition and Planarization:**
 - Deposit an **oxide or dielectric layer** over the entire structure.
 - Use **Chemical Mechanical Polishing (CMP)** to flatten the surface and expose the top of the dummy gate.

4. **Dummy Gate Removal:**
 - Etch away the polysilicon dummy gate selectively without damaging surrounding materials.
 - This leaves an empty trench where the final gate will be formed.
5. **Metal Gate Deposition:**
 - Deposit **metal layers** (e.g., TiN, Al, W) into the trench to form the final gate stack.
 - Different metals are used for NMOS and PMOS devices to achieve proper **work function tuning**.
6. **Planarization and Post-Metallization Anneal:**
 - Perform **CMP** again to remove excess metal and achieve a smooth surface.
 - Conduct a **final anneal** to improve metal-dielectric interface quality and stabilize the gate structure.

70.1923	50	43.9423	2.38502		
71.1538	50	43.6538	5.35463		
72.1154	50	43.3654	11.7874		
73.0769	50	43.0769	25.4404		
74.0385	50	42.7885	41.6426		
75	50	42.5	85.5123		
75.9615	50	42.2115	85.5123		
76.9231	50	41.9231	124.668		
77.8846	50	41.6346	242.627		
78.8462	50	41.3462	461.528		
79.8077	50	41.0577	857.565		
80.7692	50	40.7692	1035.73	Average	2599.963
81.7308	50	40.4808	1808.64		
82.6923	50	40.1923	3075.3		
83.6538	50	39.9038	3152.28		
84.6154	50	39.6154	4995.31		
85.5769	50	39.3269	7673.21		
86.5385	50	39.0385	11405.7		
87.5	50	38.75	9379.63		
88.4615	50	38.4615	12827.8		
89.4231	50	38.1731	16897.1		
90.3846	50	37.8846	11683.8		
91.3462	50	37.5962	14168.3		

Device is working as the average is 2599.963 which is leesser than 10^{12}

2)

a) What are the process parameters that affect the Replacement Gate process most?

Answer:

1. **Etch Parameters (During Dummy Gate Removal and Gate Etch):**
 - Etch rate and etch selectivity between materials (e.g., polysilicon vs. oxide) determine how accurately the dummy gate can be removed without damaging the surrounding dielectric or spacers.
 - Etch time and plasma power affect the final Gate Critical Dimension (CD) and trench profile.
 - Over-etching can lead to gate undercut or void formation, while under-etching can cause incomplete removal of the dummy gate.
2. **Deposition Parameters (Metal Fill and Dielectric Deposition):**
 - Deposition rate, step coverage, and film conformality influence how well the metal fills the narrow gate trench.
 - Poor step coverage or low deposition rate can result in voids or seam formation inside the metal gate.
 - Deposition temperature also affects film stress and metal diffusion.
3. **Lithography Parameters:**
 - Exposure dose, focus, and lithography bias determine the accuracy of the gate length (Gate CD).
 - A small variation (e.g., ± 5 nm) in lithography bias can significantly impact the threshold voltage and device performance.
 - Overlay accuracy between gate and source/drain masks is critical — misalignment can cause

leakage or drive current reduction.

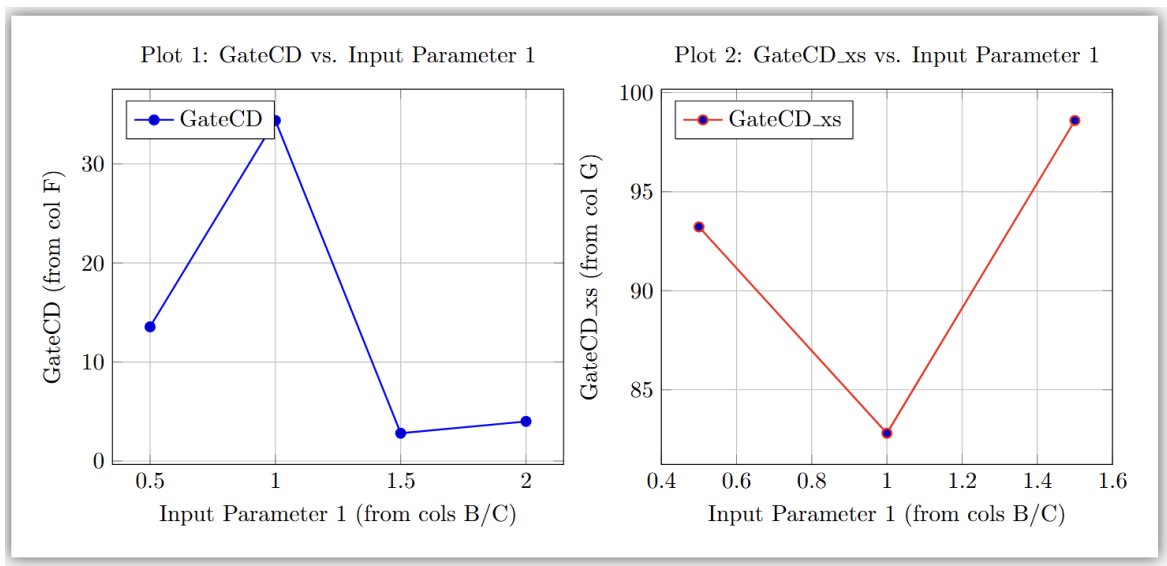
4. Annealing Conditions:

- Anneal temperature and time affect dopant activation, stress relaxation, and metal diffusion.
- Excessive temperature during the dummy anneal can cause metal contamination (e.g., Al diffusion $> 10^{12} \text{ cm}^{-3}$), leading to device failure.

b) Use virtual metrology to plot Gate CD versus Gate etch parameters.

Answer:

# Comman Expeditor V2_ReplacementGate_cd.csv -dir V2_ReplacementGate_cd_Exp -reorder -killflagfile C:/Users/Devjeet Nanda/AppData/Local/Temp/semuKillFilePPU6VL -pauseflagfile C:/Users/Devjeet Nanda/Downloads/V2_ReplacementGate/expeditor_pi														
# Started 04-11-2025 12:57														
# Output to C:/Users/Devjeet Nanda/Downloads/V2_ReplacementGate/V2_ReplacementGate_cd_Exp														
# Assembl V2_ReplacementGate.zam														
# Step Narr Multi Etch (Requires Advanced Moc Multi Etch Multi Etch Multi Etch Multi Etch (Requires Advanced Modeling)														
Run	3.6:materials[/Organics/Resist].e	3.6:materi	3.6:materi	3.6:materi	3.6:materi	GateCD	GateCD_xs	GateCD_ys	GateCD_zs	GateCD_xf	GateCD_yf	GateCD_zf	Duration	
Run-0	0.5	0.5	1	1	1	13.55872	93.220642	57.499999	57.36719	106.77935	57.499999	57.36719	0:02:27 (H:M:S)	
Run-1	0.5	0.5	1	1	1	13.55872	93.220642	57.499999	57.36719	106.77935	57.499999	57.36719	0:01:49 (H:M:S)	
Run-2	0.5	0.5	1	1	1	13.55872	93.220642	57.499999	57.36719	106.77935	57.499999	57.36719	0:01:51 (H:M:S)	
Run-3	0.5	0.5	1	1	1	13.55872	93.220642	57.499999	57.36719	106.77935	57.499999	57.36719	0:01:52 (H:M:S)	
# Finished 04-11-2025 13:02														
# Run time 0:04:49 (H:M:S)														



c) Give two examples of how to decrease the Gate CD.

Answer:

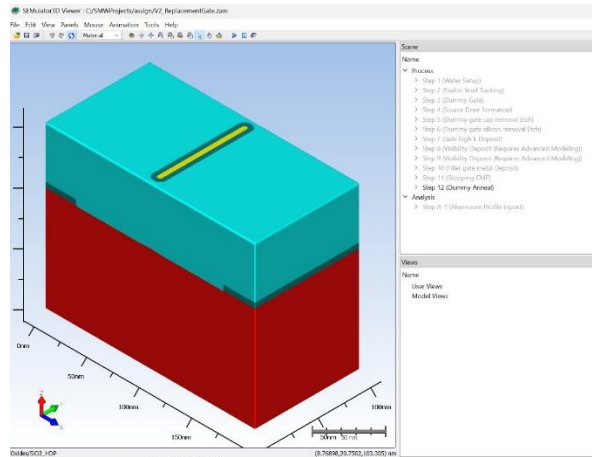
1. Increase the Etch Time or Etch Rate during Gate Etching

- By slightly increasing the etch duration or plasma power, more material is removed from the gate sidewalls. This leads to a narrower gate width (smaller CD).
- Excessive etching can cause gate undercut, profile distortion, or void formation during metal filling, so the adjustment must be precisely controlled.

2. Apply a Negative Lithography Bias (Reduce Exposure Dimension)

- During lithography, a negative bias (e.g., -5 nm) can be applied to the gate mask pattern to intentionally print a smaller gate width on the wafer. This reduces the Gate CD before etching even begins.
- Too much lithography bias can cause critical dimension variation, alignment issues, or short-channel effects in the final device.

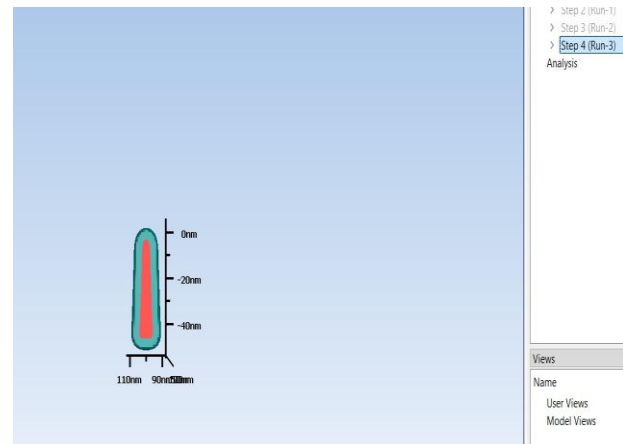
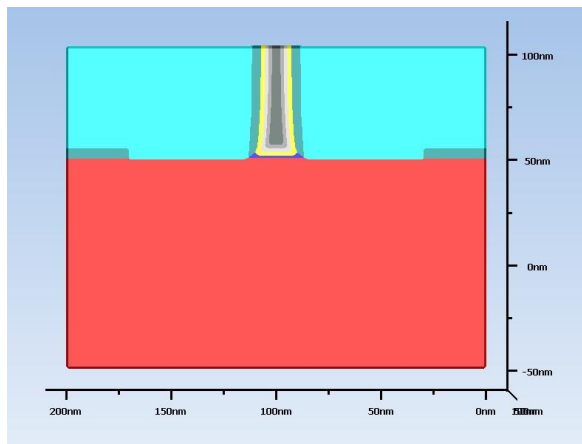
d) Go to the gate lithography step. Introduce 5nm lithography bias. Run the process. Is the device working?



70.1923	50	43.9423	0	
71.1538	50	43.6538	3.31E-10	
72.1154	50	43.3654	7.31E-10	
73.0769	50	43.0769	1.79E-09	
74.0385	50	42.7885	3.49E-09	
75	50	42.5	8.78E-09	
75.9615	50	42.2115	8.78E-09	
76.9231	50	41.9231	1.56E-08	
77.8846	50	41.6346	3.78E-08	Average 8.17489E-06
78.8462	50	41.3462	8.98E-08	
79.8077	50	41.0577	2.09E-07	
80.7692	50	40.7692	3.09E-07	
81.7308	50	40.4808	6.81E-07	
82.6923	50	40.1923	1.46E-06	
83.6538	50	39.9038	1.83E-06	
84.6154	50	39.6154	3.67E-06	
85.5769	50	39.3269	7.15E-06	
86.5385	50	39.0385	1.35E-05	
87.5	50	38.75	1.34E-05	
88.4615	50	38.4615	2.34E-05	

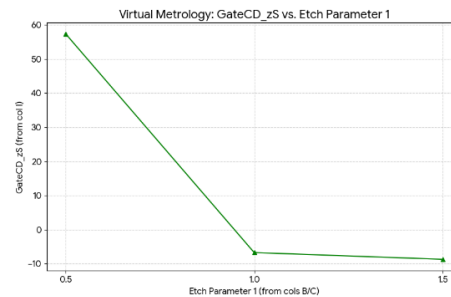
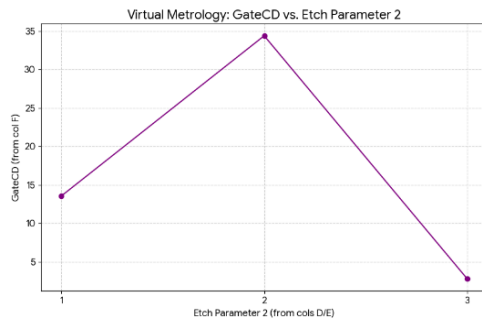
- The device is working as the average Al doping is 8.17489E-06 is below 10^{12} .

e) Undo the process change of 2.d. For what etch and/or deposition parameters Replacement Gate process fails, notice void formation.



f) Use the 'Void metrology' analysis result to measure the length, depth and area of the void.

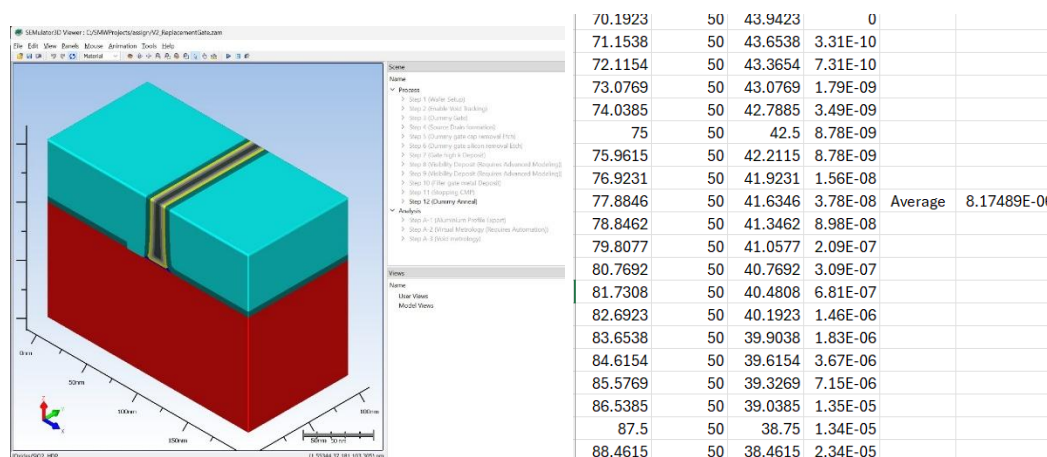
#Step Name	Wafer Setup	Void metrology	Void metrology							
Run	1:thickness	Analysis.3:	Analysis.3:	GateCD	GateCD_xS	GateCD_yS	GateCD_zS	GateCD_xE	GateCD_yE	GateCD_zE
Run-0	100	RMG_susbs	CD_measur	13.55872	93.220642	57.499999	57.36719	106.77935	57.499999	57.36719
Run-1	1	1	1							
Run-2	1.5	1.5	1.5	13.56763	93.21619	57.499999	8.34375	106.78381	57.499999	8.34375
Run-3	2	2	2	13.5459	93.22705	57.499999	8.375	106.7729	57.499999	8.375



These measurements (especially the depth) confirm that the metal deposition failed to fill the bottom of the high-aspect-ratio trench, leading to a fatal "pinch-off" void that would cause device failure.

g) Go to the lithography step when opening the Source and Drain region. Change the lithography mask with the 'Source_Drain2' mask. Introduce a 10nm overlay (in X direction) in the lithography step of the gate. Run the process. Is the device still working?

Answer:



- The device is working as the average Al doping is 8.17489E-06 is below 10^{-12} .

Conclusion:

The Gate Last process, also known as the replacement metal gate (RMG) technique, is a key advancement in modern CMOS fabrication that improves transistor performance and reliability. In this approach, the high-k dielectric is deposited first, while the metal gate is formed later in the process. This prevents high-temperature steps from degrading the metal gate materials and ensures better control over the threshold voltage. As a result, the Gate Last method provides higher drive current, reduced leakage, and enhanced scalability for advanced technology nodes below 45 nm, making it the preferred choice for high-performance and low-power integrated circuits.

